Shipment ETA Prediction & Delay Risk Estimation

Technical Report | Aiqon AI Internship Assignment | 22 October 2026

1. Executive Summary

This project implements an end-to-end machine-learning workflow to estimate a shipment's final delay, derive a predicted arrival timestamp, and show which input features are most influential to the model. Four checkpoint-specific models are evaluated: before departure, after departure, after port arrival, and after customs. The dataset is synthetically generated for this assignment; the results demonstrate the pipeline and must not be interpreted as validated performance on real shipment operations.

2. Dataset and Assumptions

The generated dataset contains 250 records across five origin–destination routes and 14 columns, including scheduled departure/arrival, planned transit duration, congestion, weather severity, departure/port/customs delays, recovery hours, final delay, and actual arrival. The exploration step reported no missing values. The target is **final_delay_hours**; the predicted arrival is calculated as scheduled arrival plus the predicted delay.

Key assumptions

- Records are simulated, not obtained from a carrier, port, customs authority, or live tracking API.
- Congestion is represented on a 1–10 scale and weather severity on a 0–5 scale.
- The generator creates final delay from simulated stage delays, weather, recovery time, and random noise; therefore, feature relationships are built into the synthetic data.
- The LOW / MEDIUM / HIGH labels are rule-based categories: less than 12 hours, 12 to less than 24 hours, and 24 hours or more. They are not calibrated probabilities.

3. Methodology

The workflow generates data, performs exploratory analysis, trains a Random Forest regressor, saves model artifacts, evaluates predictions against a mean-prediction baseline, and produces a checkpoint comparison chart. The primary model uses 200 trees, a maximum depth of 8, and a minimum of 3 samples per leaf. Evaluation uses an 80/20 random train/test split with random_state=42, giving 200 training and 50 test records.

For each checkpoint, the model uses only the intended available features: pre-departure fields and route; then departure delay; then port delay; and finally customs delay. The prediction script computes **departure_month** from scheduled_departure before inference. MAE and RMSE are reported in hours; R^2 summarizes explained variance on the held-out test split.

4. Evaluation Results

Checkpoint	MAE (h)	RMSE (h)	R^2
Before departure	5.71	7.29	0.654
After departure	5.14	6.39	0.734
After port arrival	3.47	4.47	0.870
After customs	3.12	4.05	0.893

The baseline MAE was approximately 9.52 hours at each checkpoint. The pre-departure Random Forest MAE of 5.71 hours is about 40% lower than the baseline on this split. Errors decline at later checkpoints, when additional stage-delay information is available. Since the synthetic target is generated from related stage features, this improvement is expected and should not be assumed to transfer to real shipments.

5. Exploratory Findings and Interpretability

In exploratory analysis, final delay had the strongest observed correlations with port delay (0.890), customs delay (0.667), departure delay (0.604), weather severity (0.587), and congestion (0.560). These are correlations within the simulated dataset, not causal estimates.

For the after-port-arrival model, the leading feature-importance scores printed by the demo were **port_delay_hours (0.867)**, **weather_severity (0.052)**, and **departure_delay_hours (0.026)**. Feature importance describes the model's reliance on inputs, not the percentage of a delay caused by each factor.

6. Prediction Example

For synthetic shipment ID 10 (Mumbai → Dubai), the after-port-arrival model produced a predicted delay of 17.16 hours. The scheduled arrival was 30 January 2026 at 22:00, resulting in a predicted arrival of approximately 31 January 2026 at 15:10. Under the illustrative threshold rule, this is classified as MEDIUM delay risk. This is an example from generated data, not a real shipment status or operational ETA.

7. Limitations and Next Steps

- The dataset is small and synthetic, and its feature relationships are designed by the data-generation logic.
- A single random train/test split can give an unstable estimate of generalization; repeatable cross-validation and time-based validation should be assessed.
- Risk labels are deterministic threshold bands, not calibrated probabilities. Real probability estimates would require an explicit classification/calibration approach and representative labels.
- Real deployment would require validated carrier/route history, event timestamps, missing-data handling, monitoring for drift, and evaluation by route and shipment type.
- Further work could add prediction intervals, compare with additional regressors, document data leakage checks, and evaluate performance on a temporally later test period.

8. Reproducibility

The repository contains the data-generation, exploration, training, prediction, and plotting scripts under **src/**; generated data under **data/**; serialized models under **models/**; and metrics, predictions, feature-importance outputs, and the checkpoint comparison image under **reports/**. Dependencies are listed in requirements.txt. From the project root, activate the virtual environment and run the scripts in this order:

- `source .venv/bin/activate`
- `python -m pip install -r requirements.txt`
- `python src/generate_data.py`
- `python src/explore_data.py`
- `python src/train_model.py`
- `python src/train_stage_models.py`
- `python src/predict.py`
- `python src/plot_results.py`

Conclusion: the prototype demonstrates an end-to-end ETA regression workflow and checkpoint-based feature availability. Its reported scores are useful for illustrating the approach only; operational decisions require validation with real data.

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